

LLMs as Markov processes. Zekri et al. (2024) study the equivalence between autoregressive models and general length Markov chains. Makkuva et al. (2024); Edelman et al. (2024) model sequential data as a Markov chain and analyze the properties of a single-layer transformer. Ildiz et al. (2024) establish a link between self-attention and context-conditioned Markov models. Such works generally focus on interpreting next-token prediction of a specific architecture, and do not consider the abstraction to CoT reasoning.

Metastable Markov chains. The literature on metastable Markov processes is vast (e.g., Madras and Randall, 2001; Bovier et al., 2002; Beltrán and Landim, 2011; Landim, 2018). Here we only mention the results most relevant to our theory. In particular, as reversibility is unrealistic to presume for language or reasoning models, we generally restrict our attention to works on nonreversible processes. Fritzsche et al. (2008); Jacobi (2010); Tifenchbach (2011); Fackeldey et al. (2018) study various spectral methods to identify metastable states of Markov chains. Landim (2012); Cirillo et al. (2014); Fernandez et al. (2014, 2016); Bianchi and Gaudillière (2016) analyze critical configurations and escape times for metastable dynamics, while Landim and Xu (2015) propose a recursive procedure for model reduction. Most relevant to our work, Wicks and Greenwald (2005); Betz and Le Roux (2016) give a complete hierarchical characterization of the effective dynamics of perturbed chains but only in the asymptotic limit; building on their results, we develop a new quantitative perturbation analysis throughout Appendix C.

B Preliminaries

B.1 Pseudo-Spectral Gap and Mixing

By taking Assumption 2 and multiplying ε by a constant if necessary, we assume that $c\varepsilon \leq p^\varepsilon(y|x) \leq \varepsilon$ for some $c > 0$ and all $(x, y) \in E_s$ throughout the appendix.

Definition B.1 (mixing time). For a time-homogeneous ergodic Markov chain $X = (X_t)_{t \geq 0}$ on a finite state space Ω with transition kernel p and stationary distribution π , the mixing time t_{mix} is defined as

$$t_{\text{mix}}(\epsilon) = \min \left\{ t \geq 0 : \forall s \geq t, \sup_{x \in \Omega} \|p^s(\cdot|x) - \pi\|_{\text{TV}} \leq \epsilon \right\}.$$

Definition B.2 (hitting and return times). The n th *hitting time* and *return time* of X^ε to a set $A \subseteq S$ for $n \in \mathbb{N}$ are defined as

$$\begin{aligned} \tau_{A,n}^\varepsilon &= \inf \{ t \geq 0 : |\{0 \leq t' \leq t : X_{t'}^\varepsilon \in A\}| = n \}, \\ \bar{\tau}_{A,n}^\varepsilon &= \inf \{ t > 0 : |\{0 < t' \leq t : X_{t'}^\varepsilon \in A\}| = n \}. \end{aligned}$$

In particular, we write $\tau_A^\varepsilon = \tau_{A,1}^\varepsilon$ and $\bar{\tau}_A^\varepsilon = \bar{\tau}_{A,1}^\varepsilon$. We write $\tau_x^\varepsilon = \tau_{\{x\}}^\varepsilon$, etc. for simplicity.

The chain X is *reversible* if it satisfies the *detailed balance equation*

$$\pi(x)p(y|x) = \pi(y)p(x|y) \quad \forall x, y \in \Omega.$$

While we do not assume reversibility in this paper, it is informative to compare the conditions for rapid mixing. Denote the transition matrix corresponding to p by $\mathbf{P}$ and let the eigenvalues of $\mathbf{P}$ ordered by absolute value be $1 = \lambda_1(\mathbf{P}) \geq |\lambda_2(\mathbf{P})| \geq |\lambda_3(\mathbf{P})| \geq \dots$. For reversible chains, all eigenvalues are real and the mixing time is closely governed by the (absolute) spectral gap $\gamma(\mathbf{P}) = 1 - |\lambda_2(\mathbf{P})|$ (Levin et al., 2009):

$$\frac{1}{2 \log 2\epsilon} \left(\frac{1}{\gamma(\mathbf{P})} - 1 \right) \leq t_{\text{mix}}(\epsilon) \leq \frac{1}{\gamma(\mathbf{P})} \log \frac{1}{\pi_* \epsilon},$$

where $\pi_* = \min_{x \in \Omega} \pi(x)$ is the minimum stationary probability.

For nonreversible chains, the analogous quantity to $\gamma(\mathbf{P})$ is given by the *pseudo-spectral gap*:

Definition B.3 (Paulin (2015)). The *pseudo-spectral gap* of $\mathbf{P}$ is given as

$$\gamma^\dagger(\mathbf{P}) := \max_{m \in \mathbb{N}} \frac{\gamma((\mathbf{P}^\dagger)^m \mathbf{P}^m)}{m}$$

where $\mathbf{P}^\dagger$ is the time reversal of $\mathbf{P}$, defined as $\mathbf{P}_{ij}^\dagger = \pi_j \mathbf{P}_{ji} / \pi_i$.

When X is reversible, it holds that $\gamma(\mathbf{P}) \leq \gamma^\dagger(\mathbf{P}) \leq 2\gamma(\mathbf{P})$ (Wolfer and Kontorovich, 2022, Lemma 15). Moreover, $\gamma^\dagger(\mathbf{P})$ controls the mixing time similarly to $\gamma(\mathbf{P})$:

Proposition B.4 (Paulin (2015), Proposition 3.4). For $0 < \epsilon < 1$,

$$\frac{1 - 2\epsilon}{\gamma^\dagger(\mathbf{P})} \leq t_{\text{mix}}(\epsilon) \leq \frac{1}{\gamma^\dagger(\mathbf{P})} \left(1 + 2 \log \frac{1}{2\epsilon} + \log \frac{1}{\pi_*} \right).$$

Denote the maximum row sum norm as $\|\mathbf{A}\|_{1,\infty} = \max_i \sum_j |a_{ij}|$ for $\mathbf{A} = (a_{ij})$.

Lemma B.5. For $\mathbf{A} \in \mathbb{R}^{m \times m}$ it holds that $\|\mathbf{A}\|_2 \leq \sqrt{m} \|\mathbf{A}\|_{1,\infty}$.

Proof. For arbitrary $v \in \mathbb{R}^m$ with $\|v\| = 1$,

$$\|\mathbf{A}v\|^2 = \sum_i \left(\sum_j a_{ij} v_j \right)^2 \leq \sum_i \sum_j a_{ij}^2 \leq \sum_i \left(\sum_j |a_{ij}| \right)^2 \leq m \|\mathbf{A}\|_{1,\infty}^2.$$

□

B.2 Stochastic Complementation

We denote the stochastic block matrix $\mathbf{P}^\varepsilon$ corresponding to the kernel p^ε and partition $S = \cup_{k=1}^K C_k$ as

$$\mathbf{P}^\varepsilon = \begin{pmatrix} \mathbf{P}_{11}^\varepsilon & \cdots & \mathbf{P}_{1K}^\varepsilon \\ \vdots & \ddots & \vdots \\ \mathbf{P}_{K1}^\varepsilon & \cdots & \mathbf{P}_{KK}^\varepsilon \end{pmatrix}.$$

That is, the probability $p^\varepsilon(y|x)$ is contained in the (x, y) component, and the rows of $\mathbf{P}^\varepsilon$ all sum to 1. The *stochastic complement* of $\mathbf{P}_{kk}^\varepsilon$ is defined as (Meyer, 1989)

$$\mathbf{S}_{kk}^\varepsilon = \mathbf{P}_{kk}^\varepsilon + \mathbf{P}_{k*}^\varepsilon (\mathbf{I} - \mathbf{P}_k^\varepsilon)^{-1} \mathbf{P}_{*k}^\varepsilon$$

where $\mathbf{P}_{k*}^\varepsilon$ is the k th block row of $\mathbf{P}^\varepsilon$ with $\mathbf{P}_{kk}^\varepsilon$ removed; $\mathbf{P}_{*k}^\varepsilon$ is the k th block column of $\mathbf{P}^\varepsilon$ with $\mathbf{P}_{kk}^\varepsilon$ removed; and $\mathbf{P}_k^\varepsilon$ is the principal block submatrix of $\mathbf{P}^\varepsilon$ with the k th row and column removed. When $\varepsilon = 0$, it follows that $\mathbf{P}_{ij}^0 = 0$ when $i \neq j$ and $\mathbf{S}_{kk}^0 = \mathbf{P}_{kk}^0$ is the transition matrix of p^0 restricted to C_k .

The following results are fundamental to the theory of stochastic complementation.

Theorem B.6 (Meyer (1989), Theorem 2.3). *If $\mathbf{P}^\varepsilon$ is an irreducible stochastic matrix for $\varepsilon > 0$, each stochastic complement $\mathbf{S}_{kk}^\varepsilon$ is also an irreducible stochastic matrix. Moreover, $\mathbf{S}_{kk}^\varepsilon$ is equal to the transition matrix of the reduced chain $\tilde{X}^{k,\varepsilon}$ on C_k ,*

$$\tilde{X}_t^{k,\varepsilon} := X_{\tau_{C_k,t+1}^\varepsilon}^\varepsilon, \quad t \in \mathbb{N}_0$$

obtained from X^ε by deleting transitions to states outside of C_k .

We further denote the transition kernel of $\tilde{X}_t^{k,\varepsilon}$ corresponding to $\mathbf{S}_{kk}^\varepsilon$ as s_{kk}^ε , so that $s_{kk}^0 = p^0|_{C_k}$, and its return time to a subset $A \subseteq C_k$ as $\tilde{\tau}_A^{k,\varepsilon}$.

Lemma B.7 (Meyer (1989), Theorem 6.1). *Denoting the block diagonal matrix $\mathbf{S}^\varepsilon = \text{diag } \mathbf{S}_{kk}^\varepsilon$, it holds for all $k \in [K]$ that*

$$\|\mathbf{S}_{kk}^\varepsilon - \mathbf{P}_{kk}^\varepsilon\|_{1,\infty} = \|\mathbf{P}_{k*}^\varepsilon\|_{1,\infty} \quad \text{and} \quad \|\mathbf{S}^\varepsilon - \mathbf{P}^\varepsilon\|_{1,\infty} = 2 \max_{k \in [K]} \|\mathbf{P}_{k*}^\varepsilon\|_{1,\infty}.$$

In particular, it immediately follows that

$$\|\mathbf{S}_{kk}^\varepsilon - \mathbf{P}_{kk}^\varepsilon\|_{1,\infty} \leq d_{\text{out}}\varepsilon, \quad \|\mathbf{S}^\varepsilon - \mathbf{P}^\varepsilon\|_{1,\infty} \leq 2d_{\text{out}}\varepsilon. \quad (4)$$

We now exhibit conditions on the unperturbed matrices $\mathbf{P}_{kk}^0$ which imply the bounds on the spectral gap and stationary distribution of $\mathbf{S}_{kk}^\varepsilon$ in Assumption 1 up to a constant factor. The argument can be repeated to show that Assumption 1 holds for the pretrained model $\hat{p}$ of Theorem 3.1, as the error is exponentially small. This also implies that Assumption 4 is robust to perturbation via Proposition C.8.